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A Multi-Scale Attention-Based Dense Residual Network for Sugarcane Leaf Disease Detection Using BSRI Data

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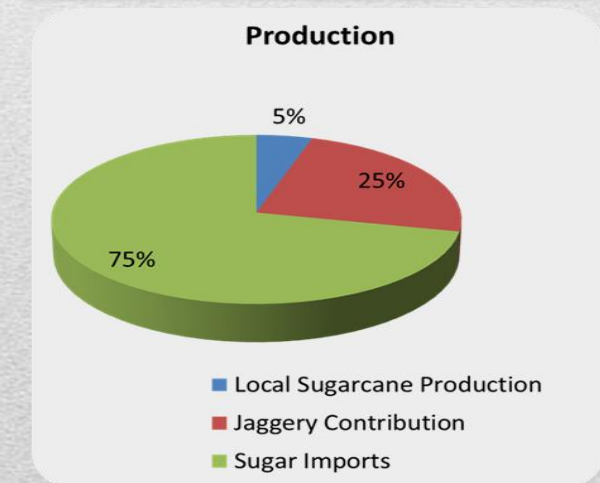
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Introduction

- Sugarcane is a vital crop for Bangladesh's agricultural economy, providing **sugar, ethanol and biofuels**.
- Bangladesh is a significant sugarcane producer in South Asia, supported by the **Bangladesh Sugarcrop Research Institute (BSRI)**, located in Ishwardi, Pabna, which focuses on sugarcane and related crop research.
- Bangladesh's modern sugar production meets only about 5% of total demand, with 20% fulfilled by jaggery production and the remaining 75% through imports [1].



Challenges Faced by Sugarcane Farmers in Bangladesh

- Sugarcane crops are highly vulnerable to fungal, bacterial, and viral diseases, **causing 30-40% yield loss** [2].
- Common leaf diseases like **yellow, rust, red rot, mosaic, and leaf scald** spread rapidly, leading to significant economic losses, crop failures, reduced sugar production, and lower product quality, affecting farmers and related industries.



Mosaic



Yellow



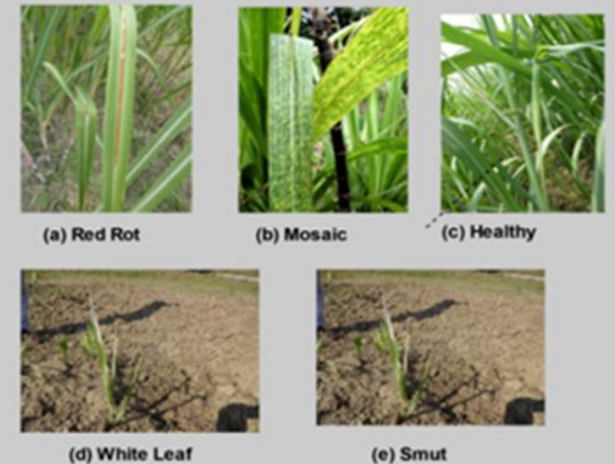
RedRot

Challenges in Developing AI-Based Disease Detection System for BSRI Dataset

- BSRI primarily focuses on analyzing the morphological characteristics of sugarcane plants, with no dedicated research on image-based disease detection.
- Only a few image samples are stored at BSRI, making them insufficient for AI-based analysis.
- There is no existing annotated image dataset; expert verification, annotation, and quality control of images are labor-intensive and time-consuming.
- Disease progression tracking requires images captured over time, which is rarely available.



Sugarcane fields in BSRI

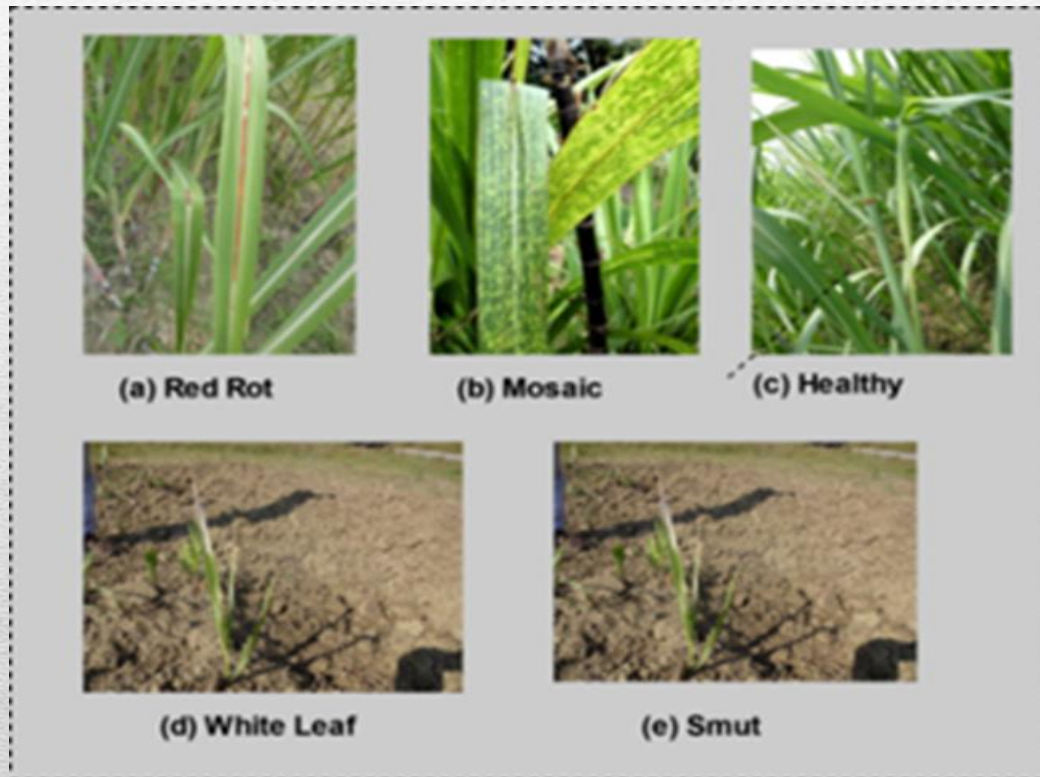


Objectives of the study

- Develop a robust and scalable system for sugarcane leaf disease detection using advanced deep learning, specifically a deep attention-based model, for effective early detection and accurate classification.
 - Improve model generalization using a combined sugarcane leaf disease dataset.
 - Build a real-time web app for disease prediction.
 - Evaluate the performance of the proposed model.
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Dataset Description

BSRI Dataset



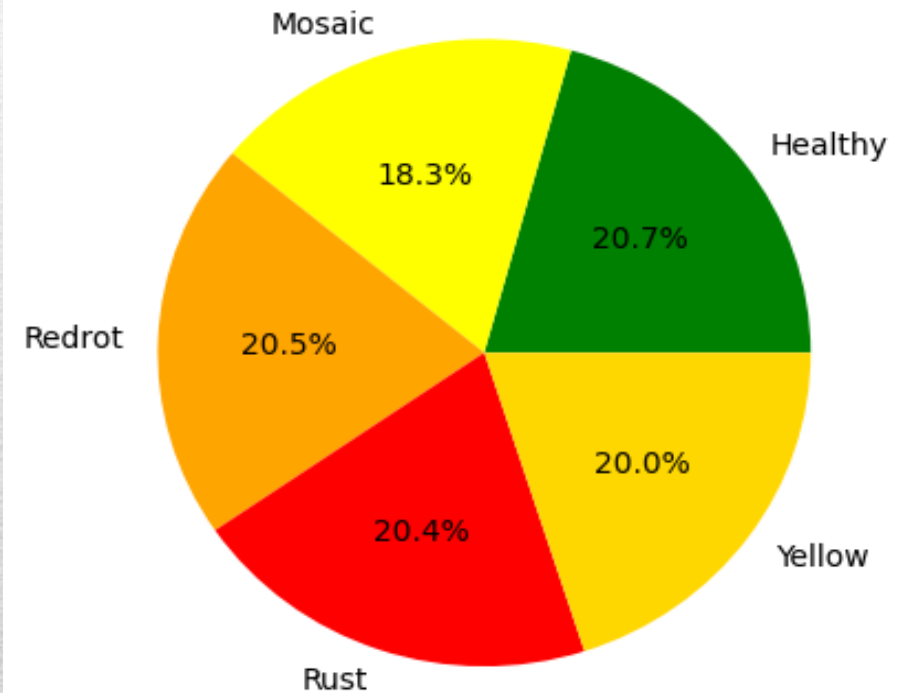
Kaggle Dataset



Dataset description

- The BSRI dataset contains 15 sugarcane leaf images categorized into five classes: healthy (3), mosaic (4), red rot (2), smut (2), and white leaf (4).
- The Kaggle dataset consists of 2521 sugarcane leaf images distributed across five classes: healthy (522), mosaic (462), redrot (518), rust (514), and yellow (504) [6]. The dataset demonstrates a well-balanced distribution, ensuring equal representation of all disease categories.

Distribution of Sugarcane Leaf Disease Images



Existing Methodology

Handcrafted Features & Convolutional Neural Network (CNN)

- Color Histograms
- Histogram of Oriented Gradients (HOG)
- LeNet, VGGNet, StridedNet [5]

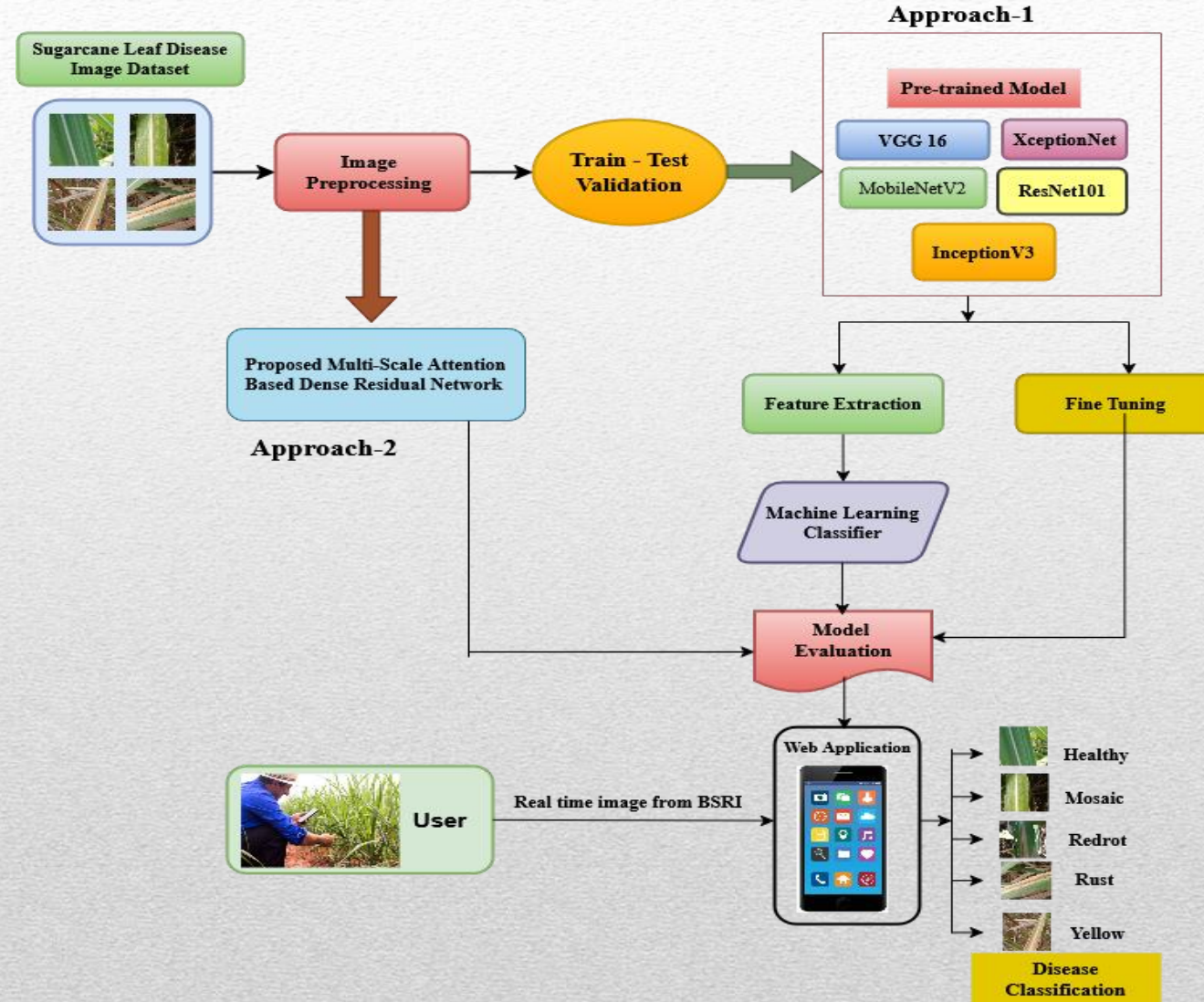
Transfer Learning Based Approach

- Feature Extractors (VGG-16, InceptionV3, VGG-19) [4]
- Fine Tuning

Attention Mechanism

- Convolutional Block Attention Module (CBAM)
 - Attention Based Multi-Level Residual Convolutional Neural Network (AMRCNN) [3]
-

Proposed Methodology Based on Attention Module





Leaf Disease Detection with Proposed Multi-Scale Attention Based Dense Residual Network (MADRN)

Attention module: what is?

- An attention module is a mechanism in deep learning that allows models to focus on the **most important parts of the input data**.
- It assigns weights to features, channels, spatial regions, or time steps, enhancing the model's ability to process **relevant information** while **ignoring less important parts**.



Attention-based Convolutional Neural Network (CNN)

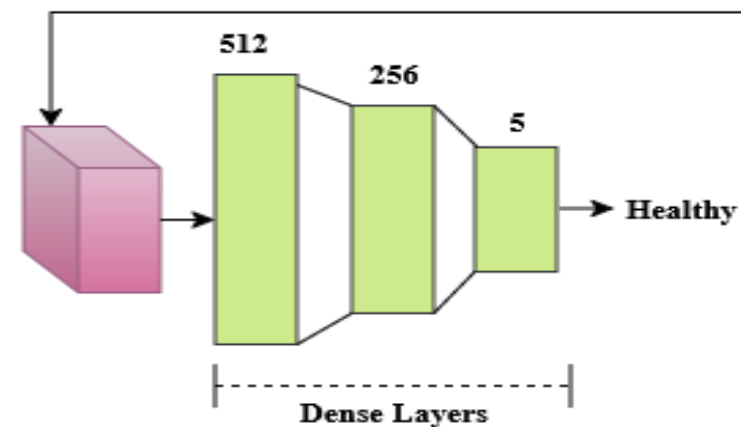
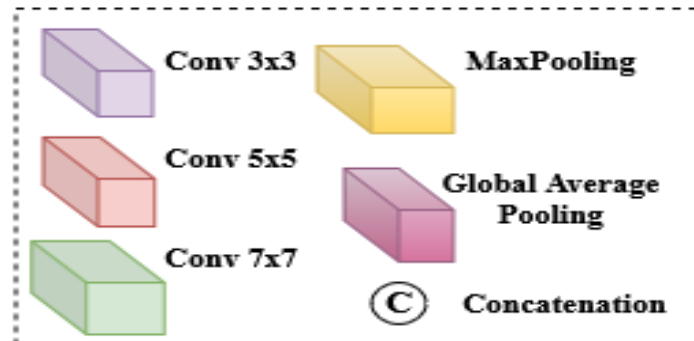
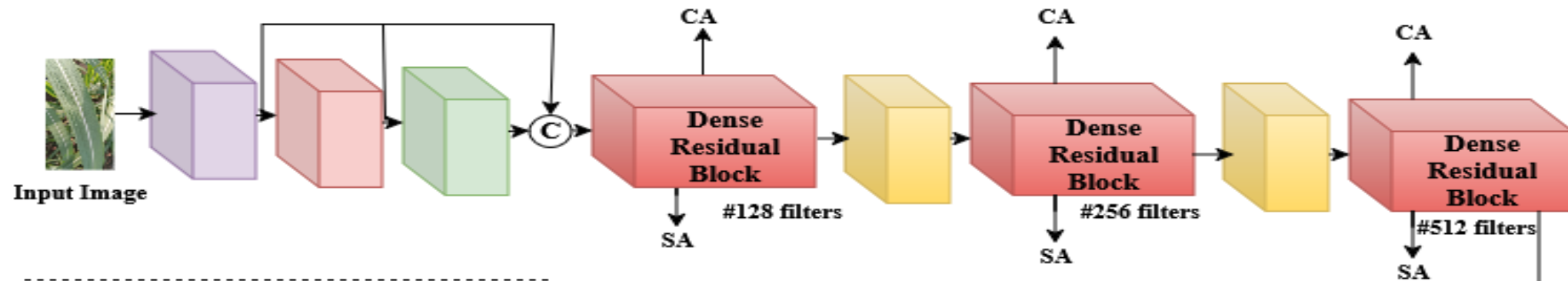
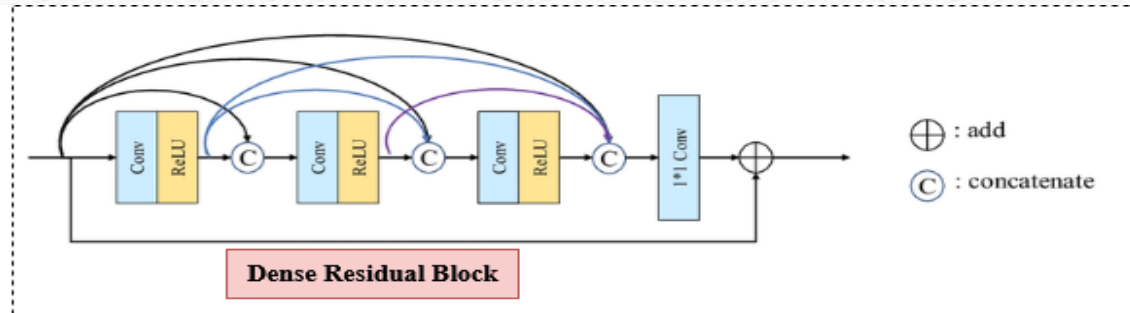
➤ **Definition:**

Combines convolutional neural networks (CNNs) with attention mechanisms to focus on important features in input data.

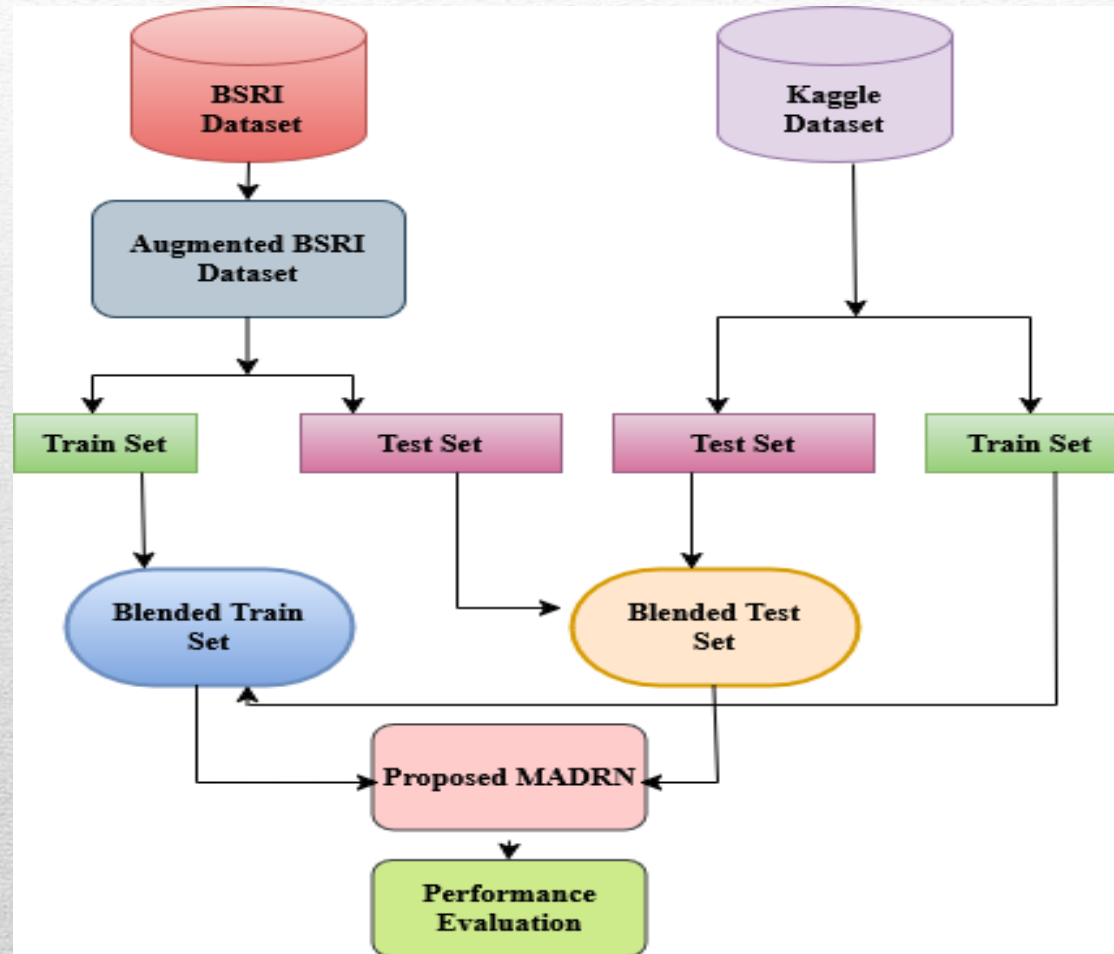
➤ **How it works:**

- CNN extracts spatial patterns (edges, textures).
 - Attention layer assigns weights to key features, enhancing relevant information while ignoring noise.
-

Proposed Multi-Scale Attention Based Dense Residual Network (MADRN)



Proposed MADRN with Blended Dataset



Results of Proposed MADRN Model

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN+CBAM	86.85	80.56	81.25	83.06
DenseNet+CBAM	83.55	79.06	80.93	81.35
ResNet+CBAM	75.20	65.76	67.69	70.77
Proposed MADRN with Kaggle Dataset	94.78	91.00	93.00	92.00
Proposed MADRN with Blended Dataset	92.25	93.00	92.00	92.00

Benchmarking With Similar Existing Studies

Author	Technique	Accuracy (%)	AUC (%)
Srivastava et al.	VGG-16, InceptionV3, VGG-19	-----	90.20
Daphal et al. [3]	VGG19, ResNet50_v2, XceptionNet, AMRCNN	86.53	94.00
Kunduracioglu et al.	InceptionV4, EfficientNet-b6	93.10	-----
Bou et al.	Integrating HIS with deep learning techniques	90.00	-----
Padila et al.	SVM	86.63	-----
Our Proposed Study	Proposed MADRN on Kaggle dataset	94.78	99.00
	Proposed MADRN on blended dataset	92.25	94.00

Web Application Development

Plant Leaf Disease Prediction

Upload an image of the plant leaf

Choose an image...



Drag and drop file here

Limit 200MB per file • JPG, JPEG

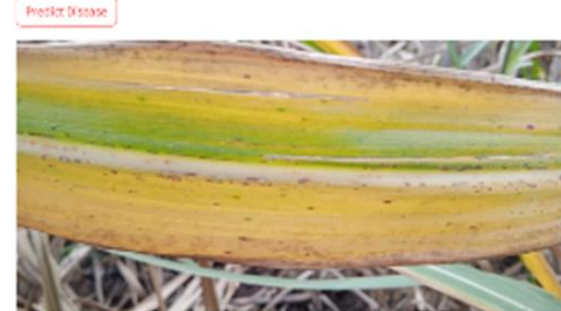
Browse files

Predict Disease



(1840, 505, 5)

This is a Mosaic leaf.

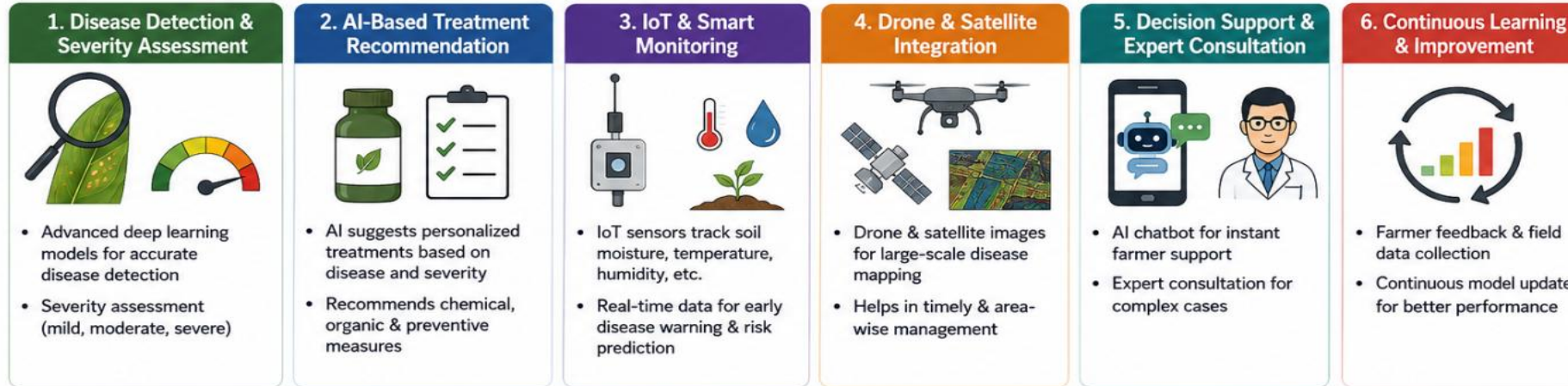


(1780, 1048, 22)

This is a Yellow leaf.

Proposed Future Extension: AI-Driven Smart Sugarcane Disease Management System

As a future extension of this research, the existing disease detection model can be expanded into a complete smart decision support system for sugarcane disease management.



Current Work (Scope of This Research)

Accurate disease detection using deep learning models.



Future Extension

Full-scale smart management system including severity assessment, treatment recommendation, monitoring, decision support & more.

Conclusion

- **Innovative Solution:** This research introduces an advanced deep learning models, including the proposed Multi-scale Attention-based Dense Residual Network (MADRN), for accurate sugarcane leaf disease detection.
 - **Significant Achievements:** The proposed approach has the potential to transform sugarcane disease management by improving productivity and promoting sustainable agricultural practices in regions where sugarcane remains a critical crop.
 - **Practical Impact:** The Web-based application and AI- powered medication recommendation system will provide a user-friendly, real-time disease detection tool, empowering farmers and agricultural professionals to make timely, data-driven decisions, enhancing crop management and reducing losses.
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References

1. —Annual-Reports - Bangladesh Sugarcrop Research Institute-Government of the People's Republic of Bangladesh. https://bsri-gov-bd.translate.google.com/site/view/annual_reports/Annual-Reports?_x_tr_sl=bn&_x_tr_tl=en&_x_tr_hl=en&_x_tr_pto=sc (accessed Jan. 17, 2025).
 2. Gonçalves, M. C., Pinto, L. R., Pimenta, R. J. G., & Da Silva, M. F. (2024). Sugarcane. *Viral Diseases Of Field And Horticultural Crops*, 193–205. <https://doi.org/10.1016/B978-0-323-90899-3.00056-2>
 3. Daphal, S. D., & Koli, S. M. (2024). Enhanced Deep Learning Technique For Sugarcane Leaf Disease Classification And Mobile Application Integration. *Heliyon*, 10(8), E29438. <https://doi.org/10.1016/J.Heliyon.2024.E29438>
 4. Huang, Y.-K., Li, W.-F., Zhang, R.-Y., & Wang, X.-Y. (2018). Diagnosis And Control Of Sugarcane Important Diseases. *Color Illustration Of Diagnosis And Control For Modern Sugarcane Diseases, Pests, And Weeds*, 1–103. https://doi.org/10.1007/978-981-13-1319-6_1
 5. Li, X., Li, X., Zhang, S., Zhang, G., Zhang, M., & Shang, H. (2023). Slvit: Shuffle-convolution-based Lightweight Vision Transformer For Effective Diagnosis Of Sugarcane Leaf Diseases. *Journal Of King Saud University - Computer And Information Sciences*, 35(6), 101401.
 6. Sugarcane Leaf Disease Dataset. <https://www.kaggle.com/datasets/nirmalsankalana/sugarcane-leaf-disease-dataset> (accessed Jan. 15, 2025).
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OPEN

Improved multiscale attention based deep learning approach for automated sugarcane leaf disease detection using BSRI data

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Early and accurate detection of sugarcane leaf diseases is critical for improving crop productivity and reducing economic losses in the agricultural sector. Timely interventions enable sustainable crop management and better resource use. In this study, we propose a deep learning-based approach for sugarcane leaf disease classification that leverages a novel architecture, the Multi-scale Attention-based Dense Residual Network (MADRN). The MADRN model integrates dense residual learning and multi-scale attention mechanisms to effectively capture fine-grained, disease-specific features and address challenges related to domain variability and complex data patterns. Two datasets are used to evaluate the model: a Kaggle dataset and a blended dataset created by combining Kaggle images with those from the Bangladesh Sugarcrop Research Institute (BSRI), simulating real-world conditions. All images undergo preprocessing steps, including resizing, normalization, and data augmentation, before training. Additionally, several baseline models (CNN, VGG16, MobileNetV2, and XceptionNet) are fine-tuned and compared with the MADRN model. Experimental results demonstrate that MADRN consistently outperforms baseline models in accuracy, precision, recall, and F1-score across both datasets, achieving up to 94.78% accuracy on the Kaggle dataset and 92.25% on the blended dataset. These findings highlight MADRN's superior ability to learn discriminative features and generalize effectively across diverse data sources, making it a promising tool for precision agriculture and disease management. To facilitate practical implementation, a web-based application is developed, enabling real-time and user-friendly disease detection. This research lays a strong foundation for the development of accurate, scalable, and practical disease classification tools that can support sustainable agricultural practices.

Keywords Sugarcane, Attention mechanism, Deep learning, Leaf disease, Image processing

Plant diseases have usually hampered plant growth and agricultural output in many regions of the globe, reducing human food supply. Among all commercially cultivated products, sugarcane holds significant economic importance in Bangladesh as one of the most profitable and widely cultivated crops across various regions. It contributes substantially to agricultural income and supports the production of key products, including sugar, bioethanol, and energy generation¹. Globally, sugarcane cultivation spans over 100 countries, including major producers like India, Brazil, Australia, and China². In Bangladesh, approximately 0.38 million acres of land are dedicated to sugarcane cultivation, yielding an annual production of nearly 5.5 million tons³. Despite its critical role in the agricultural economy, Bangladesh meets only about 5% of its total sugar demand through domestic production, with jaggery contributing 20% and the remaining 75% met through imports⁴. However, sugarcane farming in the country faces significant challenges, particularly due to the widespread occurrence of various leaf diseases. In recent decades, pest and disease infestations have adversely impacted sugarcane productivity, and rapid detection and remediation of afflicted sugarcane may greatly reduce economic damages⁵. The bulk of sugarcane disease manifestations are observed on the leaves, but due to their morphological similarities, identifying these diseases can be challenging even for expert farmers⁶. The common sugarcane leaf diseases are

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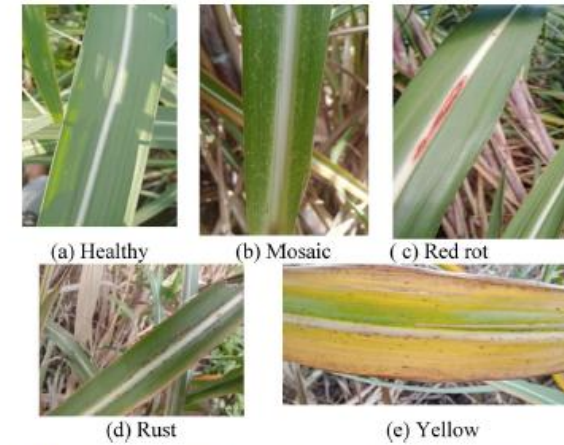


Fig. 1. Sample images representing each class from the dataset for analysis.

Class names	Image number	Percentage (%)
Healthy	522	20.71
Mosaic	462	18.33
Red rot	518	20.55
Rust	514	20.39
Yellow	505	20.02
Total	2521	100.00

Table 1. Sugarcane leaf image dataset statistic obtained from Kaggle.



Fig. 2. Sample of sugarcane leaf image dataset from BSRI.

'Yellow', 'Rust', 'Mosaic', 'Healthy'. Approximately 500 images are available in each category, ensuring balanced representation across the different conditions. Figure 1 displays a sample from each class, Table 1 and exhibits the dataset's statistical summary.

Figure 1a gives an example image from the healthy class, Fig. 1b illustrates the mosaic class, Fig. 1c represents the red rot class, Fig. 1d depicts an image from the rust class, and Fig. 1(e) depicts an image from the yellow class. In addition to the Kaggle dataset, a small dataset from the Bangladesh Sugarcane Research Institute (BSRI) has been used to test the model's adaptability to real-world conditions. This dataset includes representative images of sugarcane leaves exhibiting different disease symptoms under varied environmental conditions. The BSRI dataset covers the following disease categories: 'Mosaic', 'RedRot', 'White Leaf', 'Smut', 'Wilt', 'Healthy'. Figure 2 illustrates a sample image for every class from the BSRI dataset. The blended dataset is formed by combining images

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